

Addendum

EVALUATION

PRECISION AND RECALL - EXAMPLE



- We have a dog classifier, which, given an image, either predicts label 1=is_dog, or 0=not_dog
 - $\text{dog_clsf}(\text{image}) = c, c \in \{0,1\}$
- We want to test the performance of our classifier, to gauge whether it would perform well in the real world
- We use these 22 new (unseen) test images

PRECISION AND RECALL - EXAMPLE



- 1. Run the classifier on the test images
 - -> For these samples, our classifier said is_dog
- How well did the classifier perform?
- In reality, we have many more and complex samples, so it would be impossible to just “look” at the results
- -> Evaluate using accuracy, precision, recall, F1

PRECISION AND RECALL - EXAMPLE

Terminology

- The metrics are based on the notion of true/false positives/negatives
- True: correct classification vs. False: incorrect classification
- Positive: labelled is_dog vs. Negative: not_dog



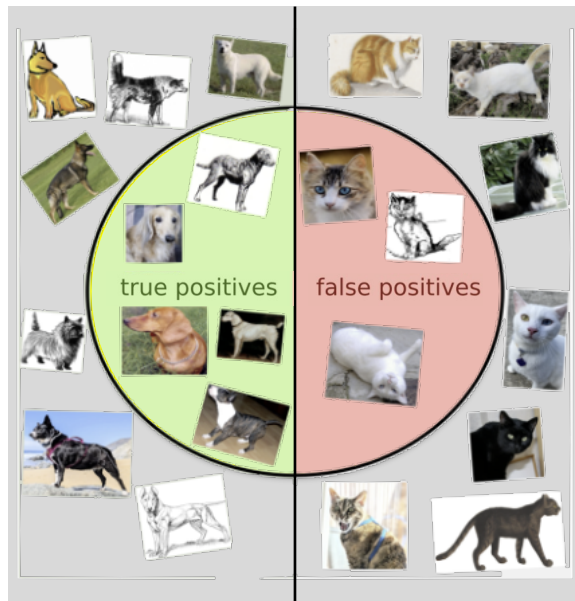
PRECISION AND RECALL - EXAMPLE

- True Positive: was labelled as `is_dog` and is actually a dog = 5

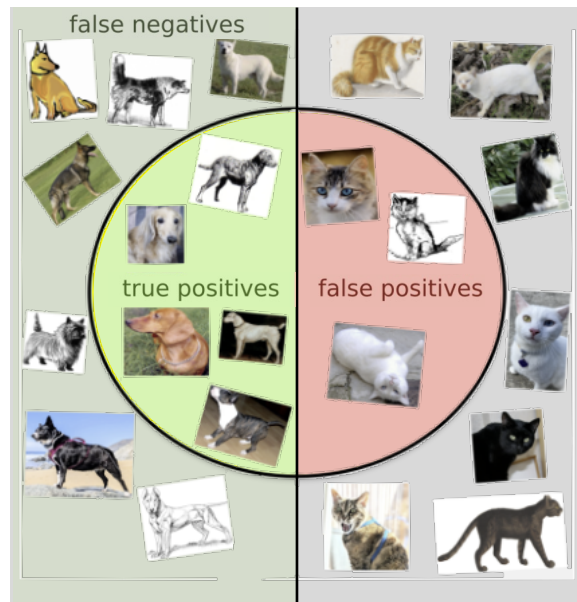


PRECISION AND RECALL - EXAMPLE

- True Positive: was labelled as `is_dog` and is actually a dog = 5
- False Positive: was labelled as `is_dog` but is actually a cat = 3

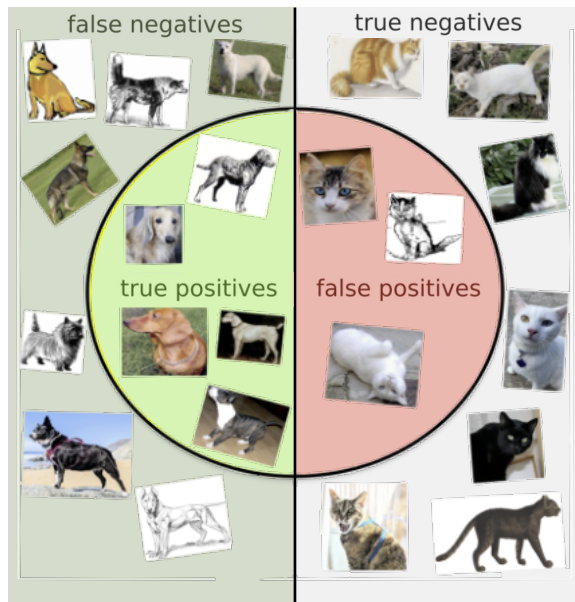


PRECISION AND RECALL - EXAMPLE



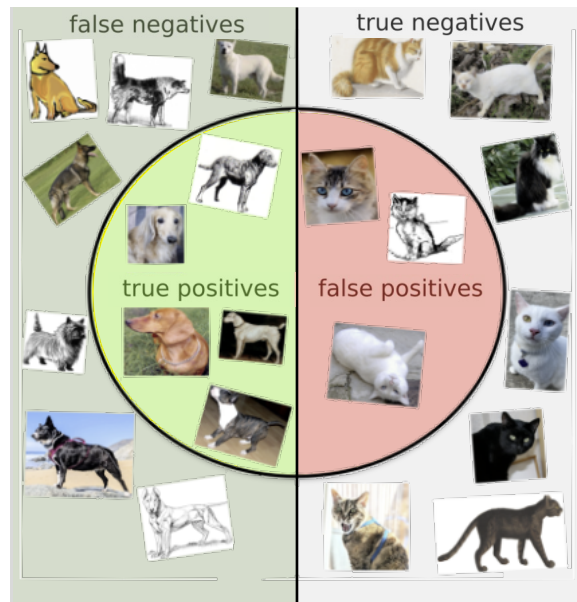
- True Positive: was labelled as `is_dog` and is actually a dog = 5
- False Positive: was labelled as `is_dog` but is actually a cat = 3
- False Negative: was labelled as `not_dog` but is actually a dog = 7

PRECISION AND RECALL - EXAMPLE



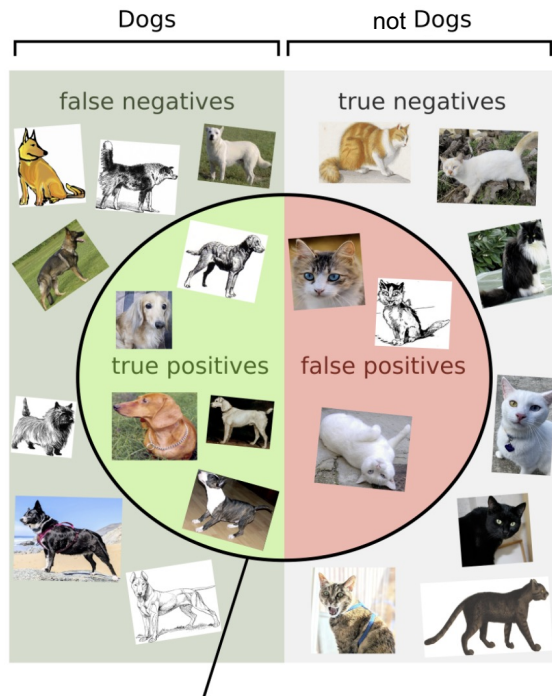
- True Positive: was labelled as `is_dog` and is actually a dog = 5
- False Positive: was labelled as `is_dog` but is actually a cat = 3
- False Negative: was labelled as `not_dog` but is actually a dog = 7
- True Negative: was labelled as `not_dog` and is not a dog = 7

PRECISION AND RECALL - EXAMPLE



- True Positive: was labelled as `is_dog` and is actually a dog = 5
- False Positive: was labelled as `is_dog` but is actually a cat = 3
- False Negative: was labelled as `not_dog` but is actually a dog = 7
- True Negative: was labelled as `not_dog` and is not a dog = 7
- In summary: TP=5, FP=3, FN=7, TN=7

PRECISION AND RECALL - EXAMPLE



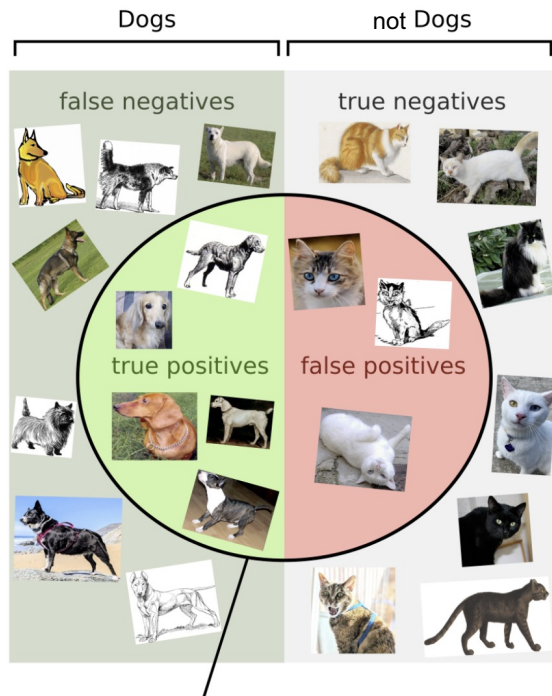
TP=5, FP=3, FN=7, TN=7

Accuracy: $\frac{TP+TN}{TP+FP+FN+TN} = \frac{5+7}{5+3+7+7} = 0.545$

“Out of all predictions, how many of our classifier’s predictions were correct?” -> 54.5%

images classified as having dogs

PRECISION AND RECALL - EXAMPLE



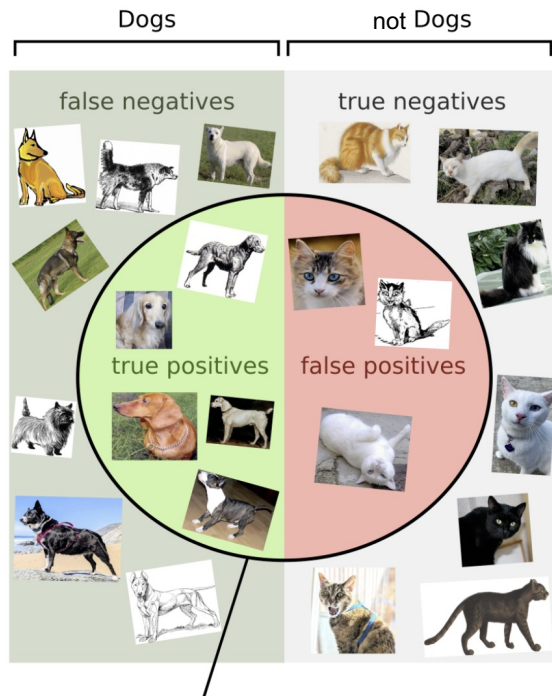
TP=5, FP=3, FN=7, TN=7

Precision: $\frac{TP}{TP+FP} = \frac{5}{5+3} = 0.625$

“Out of all is_dog predictions, how many of those were actually dogs?” / “How many retrieved items were relevant?” -> 62.5%

images classified as having dogs

PRECISION AND RECALL - EXAMPLE



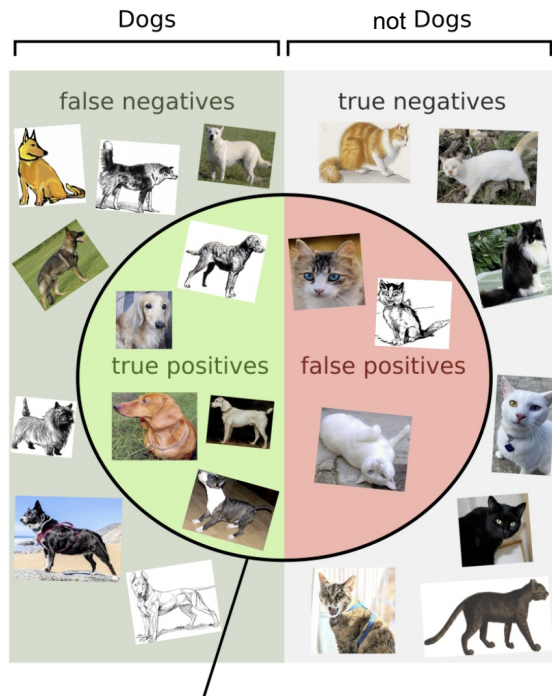
images classified as having dogs

TP=5, FP=3, FN=7, TN=7

$$\text{Recall: } \frac{5}{12} = \frac{TP}{TP+FN} = \frac{5}{5+7} = 0.417$$

“Out of all actual dogs, how many did the classifier detect?” / “How many relevant items were retrieved?” -> 41.7%

PRECISION AND RECALL - EXAMPLE



TP=5, FP=3, FN=7, TN=7

$$F1: 2 \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} = 2 \frac{0.625 * 0.417}{0.625 + 0.417} = 0.500$$

-> Harmonic mean of 50%

CONFUSION MATRIX

		Predicted condition	
		Predicted positive	Predicted negative
Actual condition	Total population = P + N		
	Real Positive (P) ^[a]	True positive (TP), hit ^[b]	False negative (FN), miss, underestimation
	Real Negative (N) ^[d]	False positive (FP), false alarm, overestimation	True negative (TN), correct rejection ^[e]

N=22	Is_dog	Not_dog
Real dog	TP = 5	FN = 7
Not dog	FP = 3	TN = 7

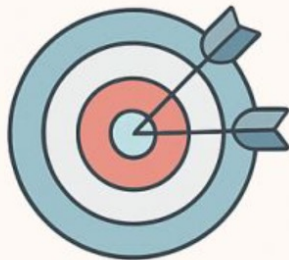
Useful to print out when doing classification, to see what the specific problems are

WHEN TO USE WHAT



ACCURACY

Use when
classes are
balanced



PRECISION

Use when
false positives
are costly



RECALL

Use when
false negatives
are costly